

# Gene flux erodes the karyotype barrier

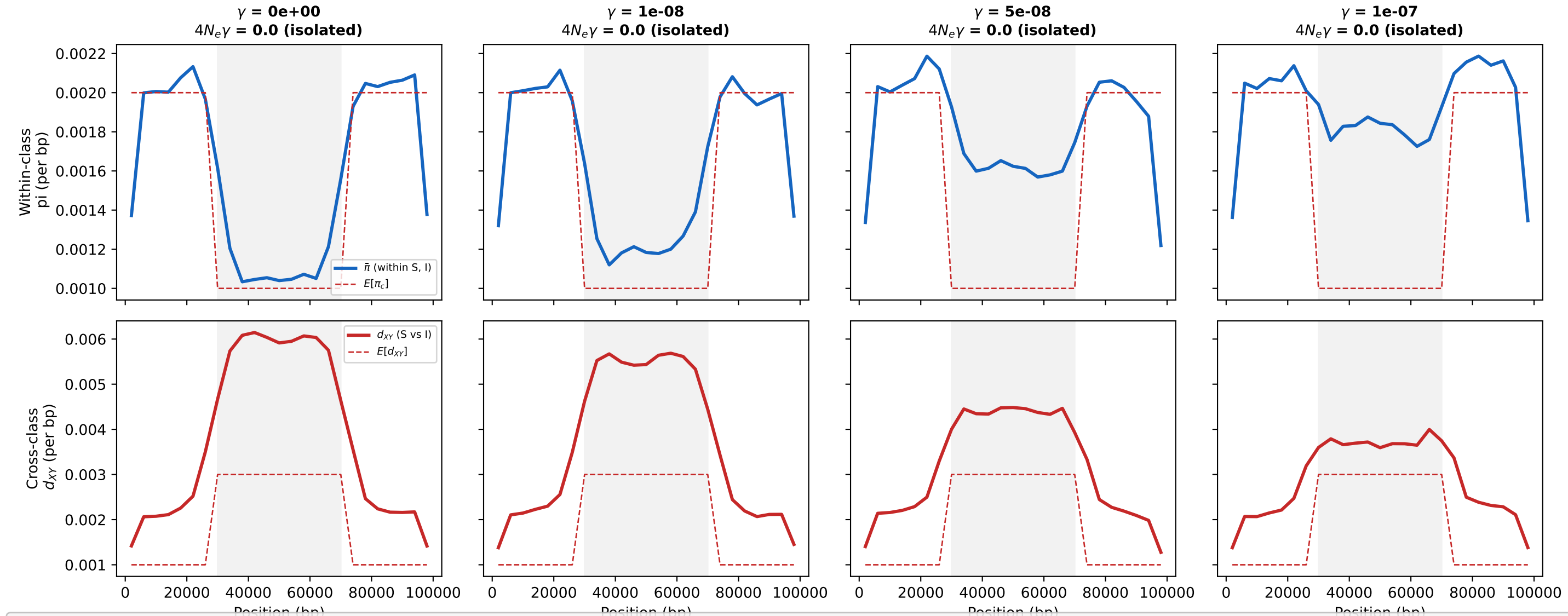


Figure. Effect of gene-conversion rate  $\gamma$  on the karyotype barrier. Each column shows a different  $\gamma$  value (0,  $1e-8$ ,  $5e-8$ ,  $1e-7$   $\text{bp}^{-1} \text{gen}^{-1}$ ). (Top) Average within-class diversity  $\bar{\pi} = (\pi_s + \pi_i)/2$ . At  $\gamma=0$  (no flux),  $\bar{\pi}$  inside the inversion (grey shading) is depressed to  $4N_e\mu \cdot p_{\text{class}}$  (red dashed) — the structured-coalescent prediction. As  $\gamma$  increases, gene flux relaxes the class barrier and  $\bar{\pi}$  recovers toward the collinear expectation  $4N_e\mu$  (grey dashed), first at the centre (where  $\phi(x)$  peaks) and last at breakpoints. (Bottom) Cross-class  $d_{xy}$  (S vs I). At  $\gamma=0$ ,  $d_{xy}$  is uniformly elevated; as  $\gamma$  grows, the centre collapses first, leaving breakpoints as the last barrier. Parameters:  $N_e=50,000$ ,  $p_{\text{inv}}=0.5$ ,  $t_{\text{inv}}=200,000$  gen,  $L=100$  kb,  $\mu=1e-8$ ,  $r=1e-8$ ,  $n_S=n_I=8$ , 40 replicates.

Command: `pixi run -e all python examples/gene_flux_demo.py`